Topic: Fourier cosine series

Question: Find the Fourier cosine series of $f(x) = \sin x$ on $0 \leq x \leq \pi$.

Answer choices:

A $$f(x) = \frac{1}{\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n + 1}{1 + n^2} \cos(nx)$$

B $$f(x) = \frac{1}{\pi} + \frac{1}{\pi} \sum_{n=2}^{\infty} \frac{(-1)^n + 1}{1 - n^2} \cos(nx)$$

C $$f(x) = \frac{2}{\pi} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n + 1}{1 + n^2} \cos(nx)$$

D $$f(x) = \frac{2}{\pi} + \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{(-1)^n + 1}{1 - n^2} \cos(nx)$$

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Solution: D

For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{\pi}\int_0^\pi \sin x\ dx$$

$$A_0 = -\frac{1}{\pi}\cos x\Big|_0^\pi$$

$$A_0 - \frac{1}{\pi}(-1-1)$$

$$A_0 = \frac{2}{}$$

And for A_n we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{\pi}\int_0^\pi \sin x\cos(nx)\ dx$$

$$A_n = \frac{1}{\pi}\int_0^\pi \sin((n+1)x) - \sin((n-1)x)\ dx$$

$$A_n = -\frac{1}{\pi}\left(\frac{1}{n+1}\cos((n+1)x) - \frac{1}{n-1}\cos((n-1)x)\right)\Big|_0^\pi$$

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$$A_n = \frac{1}{\pi}\left(\frac{1}{n+1} - \frac{1}{n-1} - \frac{\cos((n+1)\pi)}{n+1} + \frac{\cos((n-1)\pi)}{n-1}\right)$$

$$A_n = \frac{1}{\pi}\left(\frac{1-\cos((n+1)\pi)}{n+1} - \frac{1-\cos((n-1)\pi)}{n-1}\right)$$

$$A_n = \frac{1}{\pi}\left(\frac{1-(-1)^{n+1}}{n+1} - \frac{1-(-1)^{n+1}}{n-1}\right)$$

$$A_n = -\frac{1}{\pi}\left(\frac{((-1)^{n+1}-1)(n-1) - ((-1)^{n+1}-1)(n+1)}{n^2-1}\right)$$

$$A_n = -\frac{1}{\pi}\left(\frac{((-1)^{n+1}-1)(n-1-n-1)}{n^2-1}\right)$$

$$A_n = \frac{2}{\pi}\left(\frac{(-1)^n+1}{1-n^2}\right) \text{ for } n \neq 1$$

And for $n = 1$, we have

$$A_1 = \frac{2}{\pi}\int_0^{\pi} \sin x \cos x \; dx = \frac{1}{\pi}\int_0^{\pi} \sin 2x \; dx = \frac{1}{2\pi}\cos 2x\Big|_0^{\pi} = \frac{1}{2\pi}(1-1) = 0$$

Therefore, the Fourier cosine series representation of $\sin x$ on $0 \leq x \leq \pi$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2}{\pi} + \frac{2}{\pi}\sum_{n=2}^{\infty} \frac{(-1)^n+1}{1-n^2}\cos(nx)$$

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Topic: Fourier cosine series

Question: Find the Fourier cosine series of $f(x) = 2x + 5$ on $0 \leq x \leq L$.

Answer choices:

A $$f(x) = \frac{5}{2} + \frac{L}{2} - \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = 5 + L - \frac{4L}{\pi^2} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = \frac{5 + L}{2} - \frac{2L + 5}{\pi} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n} \cos\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = 5 + L - \frac{4L + 10}{\pi} \sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n} \cos\left(\frac{n\pi x}{L}\right)$$

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Solution: B

For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^L 2x + 5\ dx$$

$$A_0 = \frac{1}{L}(x^2 + 5x)\Big|_0^L$$

$$A_0 = L + 5$$

And for A_n we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{L}\int_0^L (2x + 5)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

Use integration by parts with $u = 2x + 5$, $du = 2\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ to evaluate the integral.

$$A_n = \frac{2}{L}\left(\frac{L}{n\pi}(2x + 5)\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L - \frac{2L}{n\pi}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\ dx\right)$$

$$A_n = \frac{2}{L}\left(\frac{L}{n\pi}(2L + 5)\sin(n\pi) - \frac{5L}{n\pi}\sin 0 + \frac{2L^2}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\right)\Big|_0^L$$

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$$A_n = \frac{2}{L}\left(\frac{2L^2}{(n\pi)^2}(\cos(n\pi) - 1)\right)$$

$$A_n = -\frac{4L}{(n\pi)^2}(1 - (-1)^n)$$

$$A_n = -\frac{4L}{(n\pi)^2}(1 + (-1)^{n+1})$$

Therefore, the Fourier cosine series representation of $f(x) = 2x + 5$ on $0 \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = 5 + L - \frac{4L}{\pi^2}\sum_{n=1}^{\infty}\frac{1 + (-1)^{n+1}}{n^2}\cos\left(\frac{n\pi x}{L}\right)$$

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Topic: Fourier cosine series

Question: Find the Fourier cosine series of $f(x) = (x-3)^2$ on $0 \leq x \leq \pi$.

Answer choices:

A $$f(x) = \frac{(\pi-3)^3}{9\pi} + \frac{3}{\pi} + \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{(\pi-3)(-1)^n+3}{3n^2}\cos(nx)$$

B $$f(x) = \frac{(\pi-3)^3}{9} + 3 + \frac{2}{\pi^2}\sum_{n=1}^{\infty} \frac{(\pi-3)(-1)^n+3}{3n^2}\cos(nx)$$

C $$f(x) = \frac{(\pi-3)^3}{3\pi} + \frac{9}{\pi} + \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{(\pi-3)(-1)^n+3}{n^2}\cos(nx)$$

D $$f(x) = \frac{\pi-3}{3} + 9 + \frac{2}{\pi^2}\sum_{n=1}^{\infty} \frac{(\pi-3)(-1)^n+3}{n^2}\cos(nx)$$

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Solution: C

For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{\pi}\int_0^\pi (x-3)^2\ dx$$

$$A_0 = \frac{(x-3)^3}{3\pi}\bigg|_0^\pi$$

$$A_0 = \frac{(\pi-3)^3}{3\pi} + \frac{9}{\pi}$$

And for A_n we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{\pi}\int_0^\pi (x-3)^2\cos(nx)\ dx$$

Use integration by parts with $u = (x-3)^2$, $du = 2(x-3)\ dx$, $dv = \cos(nx)\ dx$, and $v = (1/n)\sin(nx)$ to evaluate the integral.

$$A_n = \frac{2}{\pi}\left(\frac{1}{n}(x-3)^2\sin(nx)\bigg|_0^\pi - \frac{2}{n}\int_0^\pi (x-3)\sin(nx)\ dx\right)$$

$$A_n = -\frac{4}{n\pi}\int_0^\pi (x-3)\sin(nx)\ dx$$

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Use integration by parts with $u = x - 3$, $du = dx$, $dv = \sin(nx)\ dx$, and $v = -(1/n)\cos(nx)$ to evaluate the integral.

$$A_n = -\frac{4}{n\pi}\left(-\frac{1}{n}(x-3)\cos(nx)\Big|_0^\pi + \frac{1}{n}\int_0^\pi \cos(nx)\ dx\right)$$

$$A_n = -\frac{4}{n\pi}\left(-\frac{\pi-3}{n}(-1)^n - \frac{3}{n} + \frac{1}{n^2}\sin(nx)\right)\Big|_0^\pi$$

$$A_n = \frac{4}{n^2\pi}((\pi-3)(-1)^n + 3)$$

Therefore, the Fourier cosine series representation of $f(x) = (x-3)^2$ on $0 \le x \le \pi$ is

$$f(x) = \frac{(\pi-3)^3}{3\pi} + \frac{9}{\pi} + \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{(\pi-3)(-1)^n + 3}{n^2}\cos(nx)$$

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